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Multilayer Reflectionless RF Bandpass Filters With Wideband Quasi-Constant Group-Delay Responses

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Abstract—A class of reflectionless RF bandpass filters (BPFs) with wideband quasi-constant in-band group-delay response in transmission for high-data-rate digital-communication systems is presented. These BPFs are developed by exploiting multilayer wideband microstrip-to-microstrip vertical transitions and lossy multi-section microstrip lines in complementary-duplexer-based architectures. A two-port-reflectionless wideband BPF with low-in-band-variation equi-ripple-type group-delay characteristics is firstly realized by using two resistively-terminated one-section microstrip lines. To further flatten its passband group delay, its counterpart based on lossy two-section microstrip lines is then conceived to attain extremely-flat wideband group-delay pattern. Subsequently, by means of a modified wideband microstrip transition with a short-circuit-ended two-section microstrip line and a resistively-terminated two-section microstrip line, an input-reflectionless BPF with two close-to-passband transmission zeros (TZs) and wideband in-band flat group-delay profile is reported. In order to obtain a higher-order flat-group-delay wideband BPF response with more TZs, its structure shaped by two in-series-cascaded units of the previous input-absorptive BPF section with two TZs is engineered, in which different impedance values for their constituent short-circuit-ended microstrip lines are utilized. The operational foundations of all the proposed wideband flat-group-delay RF BPFs are described in detail. Furthermore, for practical-validation purposes, three 2-GHz microstrip prototypes of these reflectionless wideband RF BPFs with quasi-constant group-delay responses beyond their 3-dB absolute bandwidths (BW_s) are designed, simulated, manufactured, and characterized.

Index Terms—Absorptive filter, bandpass filter (BPF), digital communications, flat group delay, inter-symbol interference (ISI), microstrip filter, multi-functional circuit, multilayer filter, phase distortion, reflectionless filter, RF filter, transmission zero (TZ), wideband filter.

I. INTRODUCTION

BANDPASS filters (BPFs) with highly-versatile functionalities, which are fundamental RF/microwave components in the RF front-end transceiver chains of wireless-

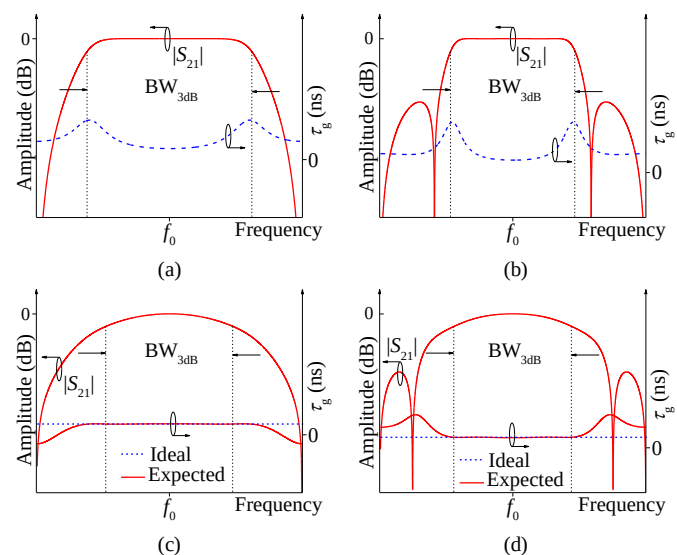


Fig. 1. Conceptual power transmission ($|S_{21}|$) and associated group-delay (τ_g) responses of typical reflective-type wideband BPFs with flat-amplitude passband and in-band concave rounded-shape group-delay profile and proposed reflectionless wideband BPFs with flat group delay beyond their 3-dB absolute BWs. (a) Typical wideband BPF. (b) Typical wideband BPF with TZs. (c) Proposed wideband flat-group-delay BPF. (d) Proposed wideband flat-group-delay BPF with TZs.

communication systems, have been explored for decades [1]. Normally, such BPFs and filtering devices are designed to exhibit flat-amplitude passbands with low insertion-loss levels and high-selectivity/sharp-rejection transfer functions through the generation of two/multiple close-to-passband transmission zeros (TZs). Nevertheless, this imposes constraints in their transmission-phase responses in terms of high in-band group-delay variations especially at the passband edges [2]. Self-equalization techniques have been engineered to enhance the in-band group-delay flatness of RF BPFs initially designed to exhibit high in-band amplitude flatness by producing complex TZs in their transfer functions through different mechanisms [3], [4]. Nevertheless, they have been mostly applied to RF narrow/moderate-bandwidth (BW) BPF designs and their practical usefulness is limited as the group-delay peaks at the transmission-band edges cannot be fully flattened or avoided. Indeed, as illustrated in Fig. 1(a) and (b), reflective-type wideband BPFs with flat-amplitude passbands typically show in-band concave rounded-shape group-delay patterns and suffer from large group-delay variations at the two passband edges. This certainly leads to undesired phase-distortion effects for in-band-processed RF signals in wireless digital-

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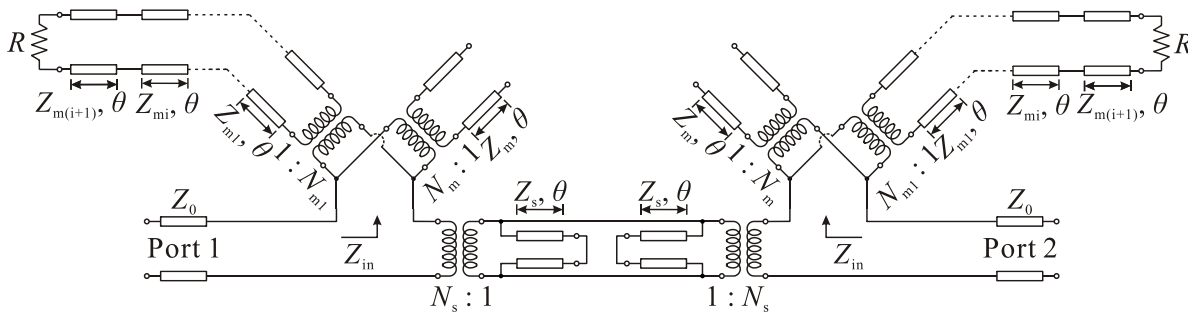


Fig. 2. Generalized TL equivalent circuit of the proposed family of two-port-reflectionless wideband flat-group-delay BPFs composed of a microstrip-to-microstrip vertical transition, two shunt resistively-terminated multi-section microstrip lines, and three pairs of transformers with turns ratios of N_m , N_s , and N_m .

communication applications [5]. As such, unwanted intersymbol interference (ISI) may arise at the detection times, which are sensitive to jitter errors for high-data-rate carriers—see Appendix A, in which the benefits derived from using flat-group-delay filters instead of more-typical filters with flatter transmission band in amplitude—but not flat group delay—are theoretically illustrated for a binary digital communications system example. Hence, it is necessary to equip RF BPFs and filtering circuits with flat and manipulable group-delay properties for their exploitation in different RF analog-signal-processing (ASP) scenarios [6], [7], [8], [9], [10], [11], [12], [13]. Based on transversal filter topologies and lossy dual-transmission structures, RF circuits with maximally-flat wideband negative group delay were reported in [8] and [9], respectively. In [10], wideband flat group-delay circuits using microstrip coupled lines were discussed. By means of acoustic-wave lumped-element resonators, BPFs and duplexers with in-band constant group-delay responses were presented in [11]. In addition, in order to circumvent the typical drawbacks of analog-to-digital conversions in ASP hardware, wideband RF filters with positive-slope linear-type-frequency-dependent group-delay profile were presented in [12]. In [13], by employing coupled-resonator circuit networks, RF dispersive-delay-structure BPFs with arbitrary asymmetric-type group-delay responses were suggested. However, all the aforementioned RF BPFs and circuits are designed as reflective-type RF devices, which means that the out-of-band/non-transmitted RF-signal power is reflected back at their input terminals.

In order to absorb the harmful stopband RF-power reflections inherent to reflective-type filters so as to enhance the stability/robustness of adjacent RF active-circuit stages, the development of their absorptive/reflectionless counterparts is becoming more attractive [14], [15], [16]. Unlike reflective-type BPFs, the non-transmitted stopband RF-power echoes can be mostly dissipated in reflectionless BPFs inside themselves by introducing lossy elements in their circuit networks, such as surface-mount-device (SMD) resistors [17], [18], [19]. In this context, reflectionless RF filters with various group-delay responses have been recently suggested as in [20], [21], [22], [23], [24], [25], [26], [27], [28], and [29]. In [20], a low-reflection microstrip BPF with narrow-band flat group-delay profile was reported. A switchable absorptive quasi-constant group-delay wideband bandstop filter (BSF) was then

presented in [21]. Moreover, by means of asymmetrical lossy stubs [22], negative-group-delay admittance inverters [23], and transversal-circuit-based power dividers [24], [25], BPFs with wideband flat-group-delay properties along with quasi or broadband absorptive behavior were engineered, respectively. Furthermore, input- and two-port-reflectionless RF lossy BSFs with static single-band, adaptive dual-band, and wideband negative-group-delay patterns were designed in [26], [27], [28], and [29], respectively. Besides, by using a two-branch-channelized circuit, a class of input-reflectionless multifunctional RF BPF with linear-type-frequency-variant in-band group-delay profile was presented in [30]. Nevertheless, most of these reflectionless RF filtering circuits are concentrated on narrow-band group-delay applications, and only some of them (e.g., [21], [22], [23], [24], [25], and [28]) are equipped with wideband group-delay responses. Furthermore, they still suffer from one or several of the following drawbacks: (i) relatively-narrow 3-dB BW along with limited reflectionless/absorptive frequency range, (ii) degraded in-band group-delay flatness, or (iii) poor filtering selectivity due to the lack of close-to-passband TZs for BPF applications.

In this paper, following the previous works in [30] and [31] on developing multilayer two-port-absorptive wideband BPFs and input-absorptive wideband sharp-rejection BPFs with remarkable passband amplitude flatness and high selectivity, respectively, a class of RF wideband BPFs with quasi-constant group-delay responses along with broadband reflectionless behavior is proposed. As illustrated in Fig. 1(c) and (d), the BPFs reported in this work are designed to exhibit wideband flat-group-delay responses beyond their corresponding 3-dB absolute-BW ranges when compared to their conceptual ideal counterparts with fully-flat group delay at the whole frequency spectrum. The main contributions of this work can be summarized as follows: (i) realization of two-port-absorptive wideband BPFs with extremely-flat in-band group-delay profile, (ii) engineering of input-reflectionless wideband BPFs with TZs and constant in-band group delay—which may allow the acquisition of broad-band RF signals with high-data-rate modulating waveforms while producing reduced ISI levels for digital communications applications—, and (iii) extrapolation to higher-order input-reflectionless wideband BPFs with flat in-band group delay and multiple pairs of TZs. To the best of the authors' knowledge, this is the first time to design multifunctional wideband RF BPFs simultaneously featuring quasi-

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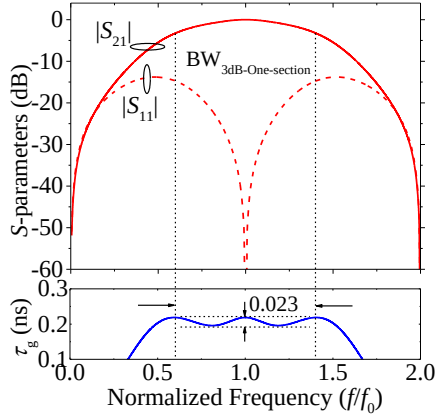


Fig. 3. Theoretical power transmission ($|S_{21}|$), input-reflection ($|S_{11}|$), and group-delay (τ_g) responses of the proposed type of wideband BPF with low-in-band-variation equi-ripple-type group delay using shunt resistively-terminated one-section microstrip lines with $z_m = 0.98$, $z_s = 1.3$, $z_{m1} = 0.8388$, and $N_m = N_s = N_{m1} = 1$.

constant in-band group-delay behavior, broadband absorptive properties, and out-of-band TZs by means of the use of wideband microstrip vertical transitions. Note that a preliminary version of this work was presented in [32], in which an example of this type of wideband flat-group-delay reflectionless RF BPF was introduced. The current study, which can be viewed as its expanded version, reports the complete theoretical background for these newly-conceived wide-band RF BPFs with absorptive capabilities and flat-group-delay behavior beyond their 3-dB BWs, along with additional proof-of-concept circuits as practical demonstrators.

The rest of this paper is organized as follows. Based on a wideband microstrip vertical transition and two shunt resistively-terminated one-section microstrip lines, two-port-absorptive wideband BPFs with low-in-band-variation equi-ripple-type group-delay responses are firstly discussed in Section II. In order to obtain extremely-flat group-delay behavior, its counterpart using shunt lossy two-section microstrip lines is then developed. Afterwards, through a modified microstrip transition with a short-circuit-ended two-section microstrip line at the output port and a shunt lossy two-section microstrip line, a type of input-reflectionless wideband BPF with a pair of TZs and flat in-band group delay is reported in Section III. By exploiting two in-series-cascaded replicas of the previously-devised input-reflectionless BPF unit with two TZs, a higher-order wideband flat-group-delay BPF with two more TZs is realized. In Section IV, a detailed comparison between the developed absorptive wideband flat-group-delay BPFs and other prior-art reflectionless wideband BPFs with quasi-constant in-band group-delay properties is made. Finally, a brief summary and the most relevant concluding remarks of this work are provided in Section V.

II. WIDEBAND FLAT-GROUP-DELAY BPFs WITH TWO-PORT REFLECTIONLESS BEHAVIOR

The generalized transmission-line (TL) equivalent circuit of the proposed family of wideband flat-group-delay BPFs with two-port-reflectionless behavior is depicted in Fig. 2. As can be seen, it consists of two main circuit parts: 1) a microstrip-

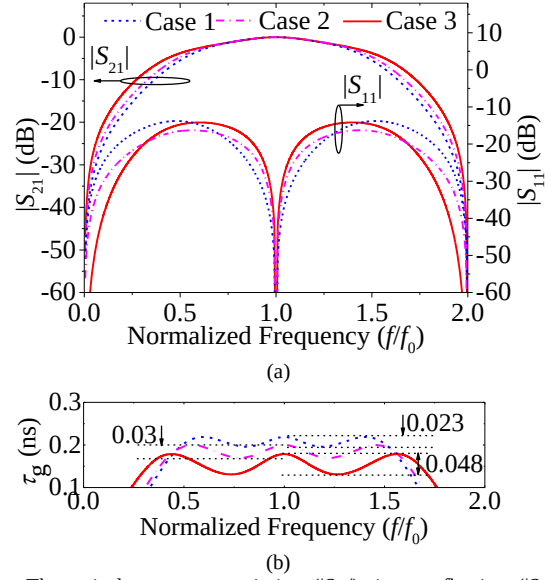


Fig. 4. Theoretical power transmission ($|S_{21}|$), input-reflection ($|S_{11}|$), and group-delay (τ_g) responses of the proposed type of wideband BPF with low-in-band-variation equi-ripple-type group delay using shunt resistively-terminated one-section microstrip lines with $N_m = N_s = N_{m1} = 1$ (Case 1: $z_m = 0.98$, $z_s = 1.3$, and $z_{m1} = 0.8388$; Case 2: $z_m = 1.006$, $z_s = 1.678$, and $z_{m1} = 0.8868$; Case 3: $z_m = 1.01$, $z_s = 2.388$, and $z_{m1} = 0.9008$). (a) $|S_{21}|$ and $|S_{11}|$ responses (Case 1: 3-dB BW of 78% from $0.61f_0$ to $1.39f_0$; Case 2: 3-dB BW of 83% from $0.585f_0$ to $1.415f_0$; Case 3: 3-dB BW of 89% from $0.555f_0$ to $1.445f_0$). (b) τ_g responses (Case 1: equi-ripple-group-delay BW of 103% from $0.485f_0$ to $1.515f_0$; Case 2: equi-ripple-group-delay BW of 117% from $0.415f_0$ to $1.585f_0$; Case 3: equi-ripple-group-delay BW of 142% from $0.29f_0$ to $1.71f_0$).

to-microstrip vertical transition for attaining a wideband BPF response [33] and 2) two shunt resistively-terminated multi-section microstrip lines for absorbing the unwanted stopband RF-power reflections. The electrical length θ of its constituent open-circuit-ended stubs, short-circuit-ended stubs, and shunt in-series-cascaded TL sections is herein set as $\pi/2$ —i.e., quarter-wavelength—at the design frequency $f_0 = 2$ GHz, and their associated characteristic impedances are Z_m , Z_s , Z_{m1} , ..., and $Z_{m(i+1)}$. Z_{in} refers to the input impedance of the shunt open-circuit-ended stub and the resistively-terminated multi-section microstrip line, Z_0 is the 50- Ω port impedance, and R represents the 50- Ω resistance of the terminating resistors that are used to attain the intended broadband two-port-reflectionless behavior. Besides, because the radiation, impedance-dispersion, and open-/short-circuit-ending effects may vary the impedance values of the coupled microstrip and slotline resonators in the practical electromagnetic (EM) design, three pairs of transformers with turns ratios of N_m , N_s , and N_{m1} are utilized [34]. In the following, the properties of the devised two-port-reflectionless wideband flat-group-delay BPFs are determined for various cases of their shunt resistively-terminated microstrip lines and numbers of their constituent in-series-cascaded sections.

A. Wideband BPFs With Low-In-Band-Variation Equi-Ripple-Type Group Delay

By considering shunt resistively-terminated one-section microstrip lines in the proposed type of two-port-absorptive BPF—i.e., $i = 0$ in Fig. 2—, its frequency responses are

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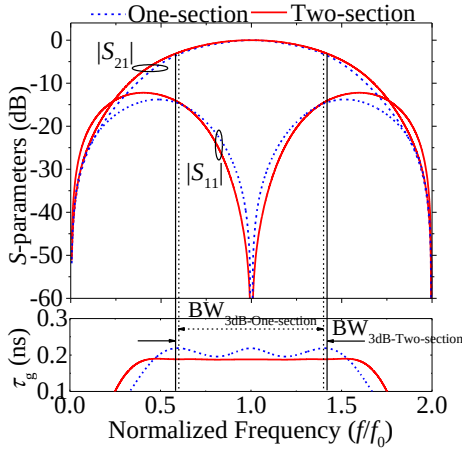


Fig. 5. Theoretical power transmission ($|S_{21}|$), input-reflection ($|S_{11}|$), and group-delay (τ_g) responses of the proposed extremely-flat-group-delay wideband BPF using shunt resistively-terminated two-section microstrip lines with $z_m = 0.768$, $z_s = 1.302$, $z_{m1} = 0.7838$, $z_{m2} = 0.8968$, and $N_m = N_s = N_{m1} = 1$ (the responses in Fig. 3 are also included for comparison purposes).

initially discussed. Through the multiplied $ABCD$ sub-matrices of the constituent elements of the whole BPF network, the analytical expressions of its power transmission (S_{21}) and reflection (S_{11}) parameters can be derived as follows:

$$S_{21} = \frac{2}{A + B/z_0 + Cz_0 + D} \quad \& \quad S_{11} = \frac{B/z_0 - Cz_0}{A + B/z_0 + Cz_0 + D} \quad (1)$$

where

$$A = D = 1 + \frac{2z'_{in}}{jz'_s \tan \theta} \quad (2)$$

$$B = \frac{2z'_{in}(z'_{in} + jz'_s \tan \theta)}{jz'_s \tan \theta} \quad (3)$$

$$C = \frac{2}{jz'_s \tan \theta} \quad (4)$$

$$z'_{in} = \frac{rz'_m z'_{m1} + jz'_m z'^2_{m1} \tan \theta}{(z'_m z'_{m1} - z'^2_{m1} \tan^2 \theta) + j(z'_m + z'_{m1})r \tan \theta} \quad (5)$$

and $z'_m = N_m^2 \times z_m$, $z'_s = N_s^2 \times z_s$, $z'_{m1} = N_{m1}^2 \times z_{m1}$, and z_0 , z_m , z_s , z_{m1} , and r are the associated normalized impedances. Based on the synthesis design method of wideband reflectionless BPFs in [31] and (1)–(5), the proposed two-port-reflectionless wideband BPF featured with quasi-equi-ripple-type passband in amplitude can be determined. Nevertheless, it exhibits an associated group-delay response with an unwanted in-band rounded shape. In order to attain maximum group-delay variations of ± 0.05 ns beyond the associated 3-dB BW ranges and broadband reflectionless behavior with power-matching levels above 10 dB within the entire frequency spectral period $[0, 2f_0)$, the design specifications of the proposed wideband BPFs corresponding to the real and imaginary parts of their S_{21} and S_{11} parameters are forced to satisfy the conditions below:

$$\Delta \tau_g = \underbrace{\frac{-\frac{1}{2\pi} d \{ \tan^{-1} [\frac{\text{Im}(S_{21})}{\text{Re}(S_{21})}] \}}_{\tau_g(f)} - \tau_g(f_0) \leq \pm 0.05 \text{ ns} \quad (6)$$

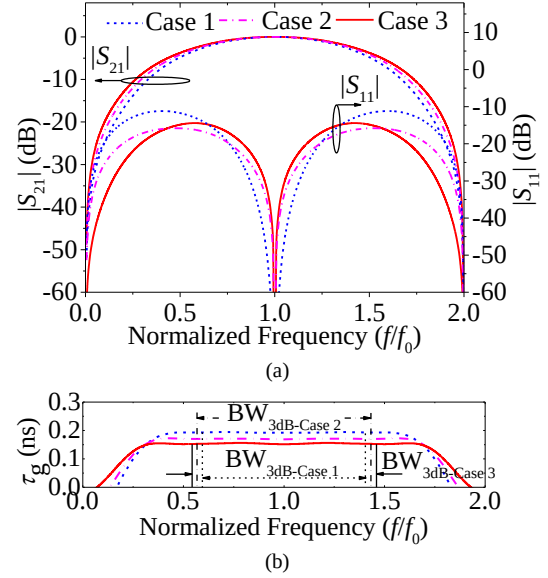


Fig. 6. Theoretical power transmission ($|S_{21}|$), input-reflection ($|S_{11}|$), and group-delay (τ_g) responses of the proposed extremely-flat-group-delay wideband BPF using shunt resistively-terminated two-section microstrip lines with $N_m = N_s = N_{m1} = 1$ (Case 1: $z_m = 0.768$, $z_s = 1.302$, $z_{m1} = 0.7838$, and $z_{m2} = 0.8968$; Case 2: $z_m = 0.807$, $z_s = 1.834$, $z_{m1} = 0.928$, and $z_{m2} = 0.897$; Case 3: $z_m = 0.797$, $z_s = 2.4$, $z_{m1} = 1.0514$, and $z_{m2} = 0.902$). (a) $|S_{21}|$ and $|S_{11}|$ responses (Case 1: 3-dB BW of 82% from $0.59f_0$ to $1.41f_0$; Case 2: 3-dB BW of 87% from $0.565f_0$ to $1.435f_0$; Case 3: 3-dB BW of 91% from $0.545f_0$ to $1.455f_0$). (b) τ_g responses (Case 1: flat-group-delay BW of 114% with maximum variation of ± 0.0024 ns from $0.43f_0$ to $1.57f_0$ and $\tau_g(f_0) = 0.1944$ ns; Case 2: flat-group-delay BW of 130% with maximum variation of ± 0.004 ns from $0.35f_0$ to $1.65f_0$ and $\tau_g(f_0) = 0.169$ ns; Case 3: flat-group-delay BW of 139% with maximum variation of ± 0.0045 ns from $0.305f_0$ to $1.695f_0$ and $\tau_g(f_0) = 0.1517$ ns).

$$|S_{11}| \leq -10 \text{ dB with } \sqrt{\text{Re}(S_{11})^2 + \text{Im}(S_{11})^2} \leq 0.3162 \quad (7)$$

Following the above design requisites with (1)–(7), the values for the impedance parameters of the devised wideband BPF using two shunt resistively-terminated one-section microstrip lines can be derived through parametric-analysis inspection. An illustrative example of its frequency responses is shown in Fig. 3. As can be seen, the designed wideband BPF features a rounded passband in amplitude with 3-dB BW range from $0.61f_0$ to $1.39f_0$ —i.e., equal to 78% in relative terms—and broadband absorptive behavior with power-matching levels above 13.78 dB within $[0, 2f_0)$. Besides, its in-band group-delay response with equi-ripple-type shape is designed with a maximum variation of -0.023 ns from $0.485f_0$ to $1.515f_0$ with regard to the group-delay value $\tau_g(f_0) = 0.2187$ ns. It results in a wideband equi-ripple-type-group-delay BW of 103% for such maximum group-delay variation, which is remarkably larger than its 3-dB-passband BW. In addition, the properties of this developed wideband BPF in terms of different group-delay ripples, as well as the associated ripple-referred group-delay BWs and 3-dB-passband BWs for distinct values of the design impedances, are illustrated in Fig. 4. In all these examples, in order to maintain the expected low-in-band-variation equi-ripple-type group-delay behavior, the values of z_m and z_{m1} were correspondingly adjusted. As shown in Fig. 4, the passband BW of the proposed BPF is increased

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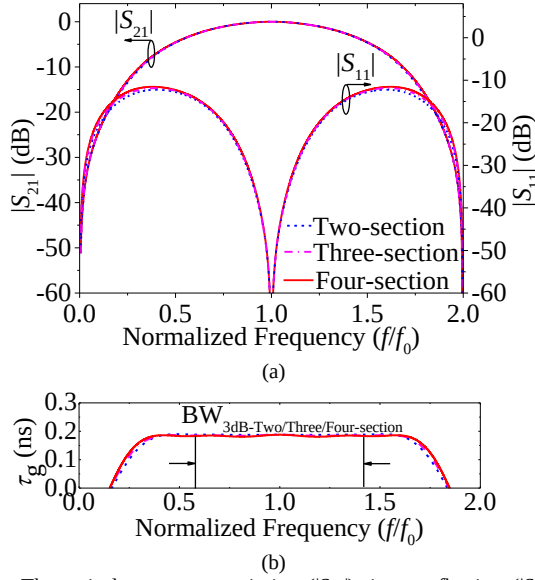


Fig. 7. Theoretical power transmission ($|S_{21}|$), input-reflection ($|S_{11}|$), and group-delay (τ_g) responses of the proposed extremely-flat-group-delay wideband BPF using shunt resistively-terminated multi-section microstrip lines for $i = 1, 2$, and 3 , respectively, with $z_m = 0.79$, $z_s = 1.402$, and $N_m = N_s = N_{m1} = 1$ (Case 1: $z_{m1} = 0.8098$ and $z_{m2} = 0.8988$ for $i = 1$; Case 2: $z_{m1} = 0.7792$, $z_{m2} = 0.8888$, and $z_{m3} = 0.9884$ for $i = 2$; Case 3: $z_{m1} = 0.7538$, $z_{m2} = 0.884$, $z_{m3} = 0.9768$, and $z_{m4} = 1.002$). (a) $|S_{21}|$ and $|S_{11}|$ responses (Cases 1, 2, and 3: 3-dB BW of 83% from $0.585f_0$ to $1.415f_0$). (b) τ_g responses (Case 1: flat-group-delay BW of 114% with maximum variation of ± 0.0018 ns from $0.415f_0$ to $1.585f_0$ and $\tau_g(f_0) = 0.1879$ ns; Case 2: flat-group-delay BW of 130% with maximum variation of -0.004 ns from $0.395f_0$ to $1.605f_0$ and $\tau_g(f_0) = 0.1879$ ns; Case 3: flat-group-delay BW of 139% with maximum variation of -0.0071 ns from $0.36f_0$ to $1.64f_0$ and $\tau_g(f_0) = 0.1879$ ns).

for a higher z_s value, which represents a larger slotline impedance of the microstrip transition [33]. Specifically, a larger 3-dB-passband BW leads to a less rounded passband, a larger ripple-referred group-delay BW, a smaller group-delay value at f_0 , but also to a higher group-delay ripple in these examples. Thus, although this BPF with shunt resistively-terminated one-section microstrip lines can attain wideband low-in-band-variation equi-ripple-type group-delay response, its maximum in-band group-delay variation needs to be further reduced as the BW is broadened.

B. Wideband BPFs With Extremely-Flat Group Delay

In order to further reduce the in-band group-delay variation so as to attain a quasi-constant-group-delay behavior beyond the 3-dB BW range, shunt resistively-terminated two-section microstrip lines in the conceived type of wideband BPF—i.e., $i = 1$ in Fig. 2—are now considered. Through the relevant equivalent circuit of this BPF in Fig. 2, the formula of its normalized input impedance is obtained to be as follows:

$$z'_{in} = \frac{r'_m z'_{m1} (z_{m2} - z'_{m1} \tan^2 \theta) + jz'_m z'_{m1} z_{m2} (z'_{m1} + z_{m2}) \tan \theta}{a_1 + jb_1} \quad (8)$$

where

$$a_1 = z'_{m1} z_{m2} (z'_m - z'_{m1}) - (z'_m + z'_{m1}) z_{m2}^2 \tan^2 \theta \quad (9)$$

$$b_1 = (z'_m z'_{m1} + z'_m z_{m2} + z'_{m1} z_{m2}) r \tan \theta - r z_{m1}^2 \tan^3 \theta \quad (10)$$

Following the design procedure discussed in Section II.A with (1)–(4), (6), (7), and (8)–(10), the theoretical frequency

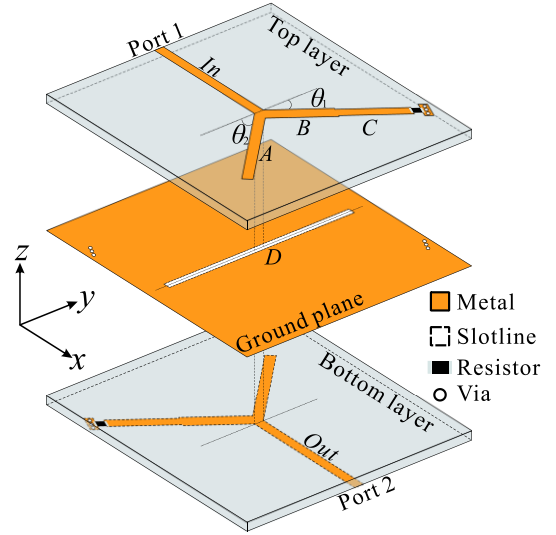


Fig. 8. 3-D layout of the developed two-port-reflectionless wideband flat-group-delay BPF prototype using shunt resistively-terminated two-section microstrip lines for $i = 1$ ($L_{in} = L_{out} = 40$, $L_A = 21.83$, $L_B = 21.65$, $L_C = 21.82$, $L_D = 49.5$, $W_{in} = W_{out} = 1.82$, $W_A = 2.43$, $W_B = 2.54$, $W_C = 2.15$, $W_D = 0.56$ [unit: mm], and $\theta_1 = \theta_2 = 45^\circ$).

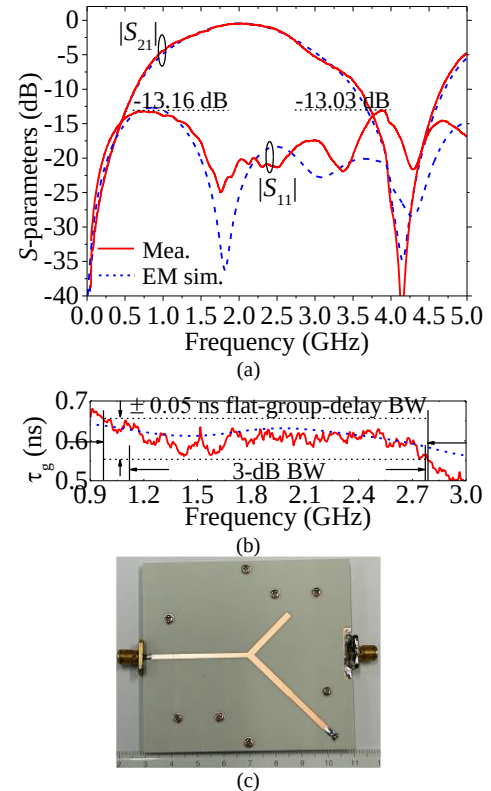


Fig. 9. Frequency responses and photograph of the manufactured microstrip BPF prototype of the conceived two-port-reflectionless wideband flat-group-delay BPF in Fig. 8. (a) EM-simulated and measured power transmission ($|S_{21}|$) and input-reflection ($|S_{11}|$) responses. (b) EM-simulated and measured group-delay (τ_g) responses. (c) Top/bottom-view photograph of the assembled prototype.

responses of an illustrative design example of the proposed BPF using resistively-terminated two-section microstrip lines are presented in Fig. 5. In comparison with the responses associated to its BPF counterpart in Fig. 3 using lossy one-section microstrip lines, a slightly-larger 3-dB BW range from

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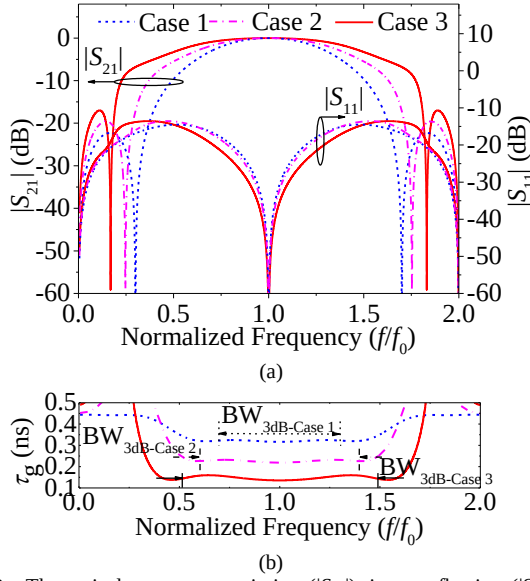


Fig. 12. Theoretical power transmission ($|S_{21}|$), input-reflection ($|S_{11}|$), and group-delay (τ_g) responses of the proposed input-reflectionless wideband flat-group-delay BPF with TZs using a shunt resistively-terminated two-section microstrip line for different z_1 values with $z_2 = 2.1614$, $z_{m1} = 1.3958$, $z_{m2} = 1.0448$, and $N_m = N_s = N_{m1} = 1$ (Case 1: $z_1 = 0.5596$, $z_m = 2.153$, and $z_s = 0.9074$; Case 2: $z_1 = 0.3596$, $z_m = 1.611$, and $z_s = 1.359$; Case 3: $z_1 = 0.1596$, $z_m = 1.101$, and $z_s = 2.229$). (a) $|S_{21}|$ and $|S_{11}|$ responses (Case 1: 3-dB BW of 60.2% from $0.699f_0$ to $1.301f_0$; Case 2: 3-dB BW of 76% from $0.62f_0$ to $1.38f_0$; Case 3: 3-dB BW of 96.8% from $0.516f_0$ to $1.484f_0$). (b) τ_g responses (Case 1: flat-group-delay BW of 85.8% with maximum variation of 0.0085 ns from $0.571f_0$ to $1.429f_0$ and $\tau_g(f_0) = 0.3163$ ns; Case 2: flat-group-delay BW of 91% with maximum variation of 0.0133 ns from $0.545f_0$ to $1.455f_0$ and $\tau_g(f_0) = 0.2189$ ns; Case 3: flat-group-delay BW of 121.4% with maximum variation of 0.0243 ns from $0.393f_0$ to $1.607f_0$ and $\tau_g(f_0) = 0.1356$ ns).

2.785 GHz—i.e., 84.76% in relative terms—, minimum in-band insertion-loss and input-power-matching levels of 0.42 dB and 13.16 dB, respectively, and input-power-matching levels above 13.03 dB from DC to 5 GHz. The simulated and measured in-band group-delay responses are depicted in Fig. 9(b). The measured group-delay variation with regard to the average in-band group-delay value of 0.6084 ns is within ± 0.05 ns from 0.972 to 2.797 GHz—i.e., 96.84% in relative terms—, which fully covers the measured 3-dB-BW frequency range. Besides, Fig. 9(c) shows the top-/bottom-view photograph of the assembled BPF prototype, which has a physical size of 80×72.46 mm².

III. WIDEBAND FLAT-GROUP-DELAY BPFs WITH TZs AND INPUT-REFLECTIONLESS BEHAVIOR

Based on the design approach of wideband two-port-absorptive quasi-constant-group-delay BPFs in Section II, the extrapolation to input-reflectionless wideband flat-group-delay BPFs with TZs using a shunt resistively-terminated two-section microstrip line is addressed here. Unlike in the two-port-absorptive BPFs in Section II, the BPF channel in the devised input-absorptive BPFs is made up of a modified wideband microstrip transition with a short-circuit-ended two-section microstrip line loaded at the output port, whereas only one shunt lossy microstrip line is employed. Its generalized

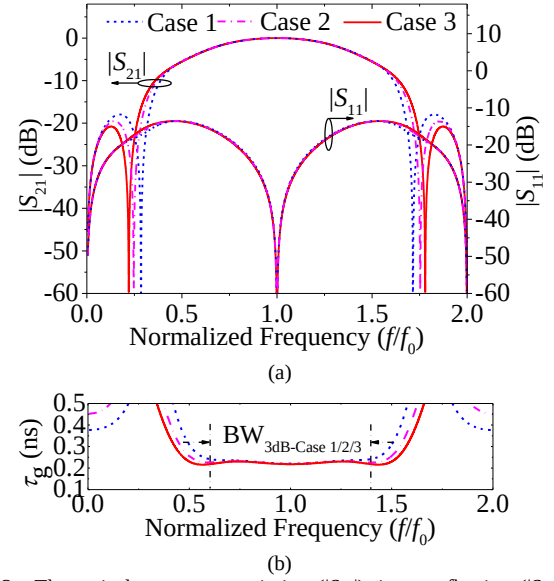


Fig. 13. Theoretical power transmission ($|S_{21}|$), input-reflection ($|S_{11}|$), and group-delay (τ_g) responses of the proposed input-reflectionless wideband flat-group-delay BPF with TZs using a shunt resistively-terminated two-section microstrip line for different z_2 values with $z_m = 1.611$, $z_s = 1.359$, $z_1 = 0.3596$, $z_{m1} = 1.3958$, $z_{m2} = 1.0448$, and $N_m = N_s = N_{m1} = 1$ (Case 1: $z_2 = 1.5614$; Case 2: $z_2 = 2.1614$; Case 3: $z_2 = 2.7614$). (a) $|S_{21}|$ and $|S_{11}|$ responses (Cases 1, 2, and 3: 3-dB BW of around 76% from $0.62f_0$ to $1.38f_0$). (b) τ_g responses (Case 1: flat-group-delay BW of 82% with maximum variation of 0.0237 ns from $0.59f_0$ to $1.41f_0$ and $\tau_g(f_0) = 0.2203$ ns; Case 2: flat-group-delay BW of 91% with maximum variation of 0.0133 ns from $0.545f_0$ to $1.455f_0$ and $\tau_g(f_0) = 0.2189$ ns; Case 3: flat-group-delay BW of 100% with maximum variation of 0.0124 ns from $0.5f_0$ to $1.5f_0$ and $\tau_g(f_0) = 0.2181$ ns).

equivalent circuit using a resistively-terminated multi-section microstrip line is depicted in Fig. 10. Similarly, the electrical length of all the TL circuit elements—i.e., open-circuit-ended stub, short-circuit-ended stubs and two-section TL, and shunt in-series-cascaded TL sections—is assumed as $\theta = \pi/2$ —i.e., quarter-wavelength—at $f_0 = 2$ GHz, and their associated line-impedance parameters are designated as Z_m , Z_s , Z_1 , Z_2 , Z_{m1} , ..., and $Z_{m(i+1)}$, respectively. The input impedances of the shunt-connected open-circuit-ended stub and lossy in-series-cascaded TL sections and the short-circuit-ended two-section TL are referred to as Z_{in} and Z_{in1} , respectively. Moreover, compared with the equivalent circuit in Fig. 2 of the two-port-reflectionless wideband BPFs, an additional transformer with turns ratio of N_{m2} is used to model the impedance variations of the short-circuit-ended two-section TL at the output port.

A. Wideband BPFs With TZs and Flat Group Delay

Following the same design procedure as for the two-port-reflectionless BPFs reported in Section II, the $ABCD$ matrix corresponding to the generalized TL equivalent circuit of the proposed type of input-absorptive BPFs with TZs using a shunt resistively-terminated two-section microstrip line—i.e., $i = 1$ in Fig. 10—can be formulated as below:

$$A = 1 + \frac{2z'_{in}}{jz'_s \tan \theta} \quad (11)$$

$$B = \frac{z'_{in}(2z'_{in1} + jz'_s \tan \theta)}{jz'_s \tan \theta} + z'_{in1} \quad (12)$$

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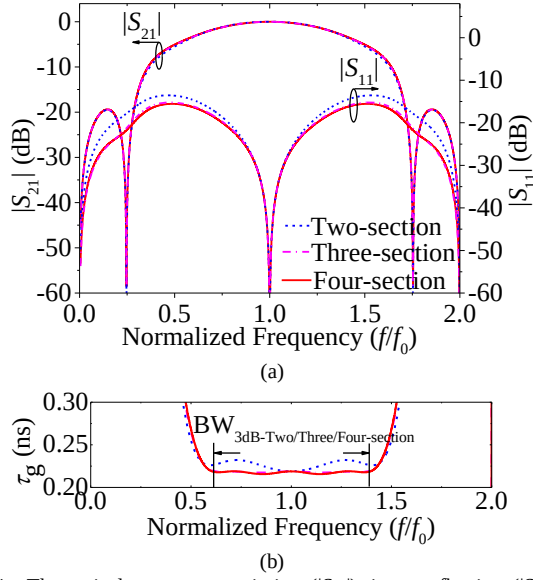


Fig. 14. Theoretical power transmission ($|S_{21}|$), input-reflection ($|S_{11}|$), and group-delay (τ_g) responses of the proposed input-reflectionless wideband flat-group-delay BPF with TZs using a shunt resistively-terminated multi-section microstrip lines for $i = 1, 2$, and 3 , respectively, with $z_m = 1.611$, $z_s = 1.359$, $z_1 = 0.3596$, $z_2 = 2.1614$, and $N_m = N_s = N_{m1} = 1$ (Case 1: $z_{m1} = 1.3958$ and $z_{m2} = 1.0448$ for $i = 1$; Case 2: $z_{m1} = 1.2508$, $z_{m2} = 1.0248$, and $z_{m3} = 0.9848$ for $i = 2$; Case 3: $z_{m1} = 1.2408$, $z_{m2} = 1.0378$, $z_{m3} = 0.9898$, and $z_{m4} = 1.009$ for $i = 3$). (a) $|S_{21}|$ and $|S_{11}|$ responses (Case 1: 3-dB BW of 76% from $0.62f_0$ to $1.38f_0$; Case 2: 3-dB BW of 77% from $0.615f_0$ to $1.385f_0$; Case 3: 3-dB BW of 77.2% from $0.614f_0$ to $1.386f_0$). (b) τ_g responses (Case 1: flat-group-delay BW of 91% with maximum variation of 0.0133 ns from $0.545f_0$ to $1.455f_0$ and $\tau_g(f_0) = 0.2189$ ns; Case 2: flat-group-delay BW of 80.6% with maximum variation of 0.015 ns from $0.597f_0$ to $1.403f_0$ and $\tau_g(f_0) = 0.2189$ ns; Case 3: flat-group-delay BW of 83.6% with maximum variation of 0.032 ns from $0.582f_0$ to $1.418f_0$ and $\tau_g(f_0) = 0.2189$ ns).

$$C = \frac{2}{jz'_s \tan \theta} \quad (13)$$

$$D = 1 + \frac{2z'_{in1}}{jz'_s \tan \theta} \quad (14)$$

$$z'_{in1} = \frac{jz'_1(z'_1 + z'_2) \tan \theta}{z'_1 - z'_2 \tan^2 \theta} \quad (15)$$

where z'_{in} has been derived as in (8)–(10), and $z'_m = N_m^2 \times z_m$, $z'_s = N_s^2 \times z_s$, $z'_{m1} = N_{m1}^2 \times z_{m1}$, $z'_1 = N_{m2}^2 \times z_1$, and z_m , z_s , z_1 , z_2 , z_{m1} , and z_{m2} are the normalized design impedances. With (1), (8)–(10), and (11)–(15), the frequency responses of this BPF network can be determined. Regarding the TZ generation condition that imposes that $S_{21} = 0$, it results in two intrinsic TZs appearing at 0 and $2f_0$ and two additional close-to-passband TZs within the $[0, 2f_0]$ spectral range. It leads to an overall BPF profile with a pair of adjacent-to-passband TZs that are produced by the used short-circuit-ended two-section TL, whose spectral locations are deduced to be as follows:

$$f_{TZ1} = \frac{2f_0}{\pi} \tan^{-1} \sqrt{k} \quad (16)$$

$$f_{TZ2} = 2f_0 \left(1 - \frac{1}{\pi} \tan^{-1} \sqrt{k}\right) \quad (17)$$

where $k = z_1/z_2$ is the impedance ratio. In this manner, in order to realize the intended input-absorptive BPF with TZs and

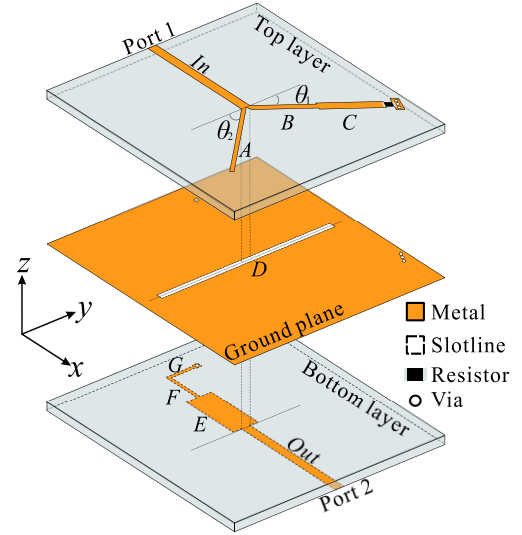


Fig. 15. 3-D layout of the developed input-reflectionless wideband flat-group-delay BPF prototype with TZs using a shunt resistively-terminated two-section microstrip line for $i = 1$ ($L_{in} = L_{out} = 40$, $L_A = 22.14$, $L_B = 22.09$, $L_C = 21.59$, $L_D = 49.7$, $L_E = 19.85$, $L_F = 13.38$, $L_G = 10.015$, $W_{in} = W_{out} = 1.82$, $W_A = 0.72$, $W_B = 1.22$, $W_C = 1.65$, $W_D = 0.48$, $W_E = 7.24$, $W_F = W_G = 0.3$ [unit: mm]), and $\theta_1 = \theta_2 = 45^\circ$).

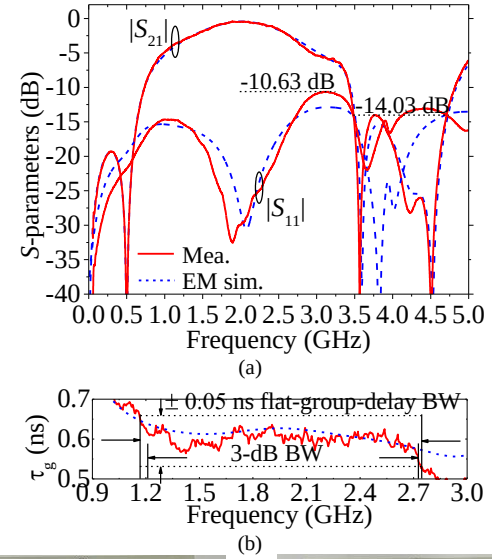


Fig. 16. Frequency responses and photographs of the manufactured microstrip prototype of the conceived input-reflectionless wideband flat-group-delay BPF with TZs in Fig. 15. (a) EM-simulated and measured power transmission ($|S_{21}|$) and input-reflection ($|S_{11}|$) responses. (b) EM-simulated and measured group-delay (τ_g) responses. (c) Top- and bottom-view photographs of the assembled prototype.

wideband in-band flat group delay, its frequency responses associated with (1), (8)–(10), (11)–(15), and a specified $k = 0.1664$ with $z_1 = 0.3596$ and $z_2 = 2.1614$ are determined to

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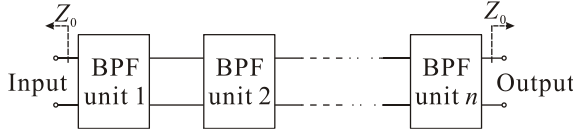


Fig. 17. Block diagram of the proposed input-reflectionless multi-TZ BPF network based on the in-series-cascaded connection of n BPF units with one pair of TZs using shunt resistively-terminated two-section TLs as in Fig. 10.

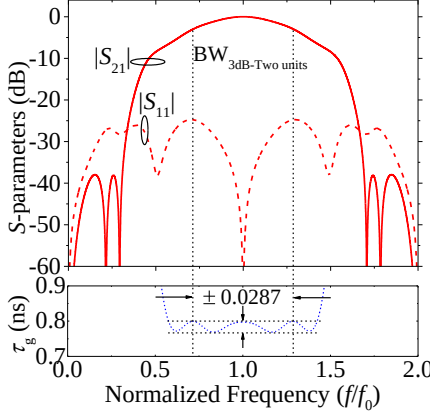


Fig. 18. Theoretical power transmission ($|S_{21}|$), input-reflection ($|S_{11}|$), and group-delay (τ_g) responses of the proposed high-order input-reflectionless wideband flat-group-delay multi-TZ BPF based on two in-series-cascaded BPF units as in Fig. 10 using shunt resistively-terminated two-section microstrip lines for different values of design impedances for $z_1 = 0.2046$ and $z_2 = 1.6614$ (unit 1) and $z_1 = 0.4228$ and $z_2 = 1.72$ (unit 2) with $z_m = 1.406$, $z_s = 1.254$, $z_{m1} = 0.9648$, $z_{m2} = 0.9938$, and $N_m = N_s = N_{m1} = N_{m2} = 1$.

satisfy the design specifications imposed in (6) and (7). As shown in Fig. 11 in which these frequency responses are depicted, a wideband BPF with a rounded sharp-rejection passband is obtained with a 3-dB BW range from $0.62f_0$ to $1.38f_0$ —i.e., 76% in relative terms—, two close-to-passband TZs at $0.247f_0$ and $1.753f_0$, input-power-matching levels above 13.544 dB, and stopband power-rejection levels above 19.503 dB within $[0, 2f_0)$. Besides, its wideband flat-group-delay curve exhibits a maximum in-band variation of 0.0133 ns from $0.545f_0$ to $1.455f_0$ —i.e., 91% in relative terms—with regard to the group-delay value $\tau_g(f_0) = 0.2189$ ns, which covers the relevant 3-dB-passband range and meets the design rules in (6).

In order to investigate adjustment flexibility in terms of 3-dB-passband BW while preserving the wideband flat-group-delay properties of the conceived input-absorptive BPF with TZs, examples of its frequency responses for different z_1 and z_2 values are drawn in Figs. 12 and 13. As shown in Fig. 12, for lower z_1 values and properly-adjusted lower z_m and higher z_s values, a larger 3-dB-passband BW with a less-rounded shape and wider associated flat group-delay range is attained. Nevertheless, slightly-deteriorated input-power-matching and stopband-power-attenuation levels are observed at the same time when the z_2 , z_{m1} , and z_{m2} values are fixed. It is also found that the flat-group-delay ranges of all these responses fully cover their associated 3-dB-BW passbands, whereas the obtained maximum in-band group-delay variation is gradually increased versus a decreased $\tau_g(f_0)$ value. On the other hand, as revealed by Fig. 13, the 3-dB-passband BW and the input-power-matching levels remain nearly unchanged as z_2 gets

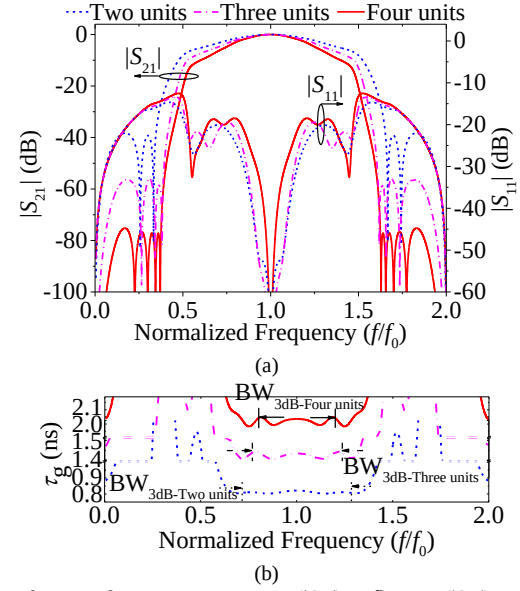


Fig. 19. Theoretical power transmission ($|S_{21}|$), reflection ($|S_{11}|$), and group-delay (τ_g) responses of the proposed high-order input-reflectionless wideband flat-group-delay multi-TZ BPFs based on in-series-cascaded BPF units as in Fig. 10 for $n = 2, 3$, and 4 , respectively, with $z_m = 1.282$, $z_s = 1.1042$, $z_1 = 0.8456$, $z_2 = 0.9668$, and $N_m = N_s = N_{m1} = 1$ (Case 1: $z_1 = 0.3612$ and $z_2 = 1.9756$ for unit 1 and $z_1 = 0.3948$ and $z_2 = 1.1884$ for unit 2; Case 2: $z_1 = 0.4652$ and $z_2 = 2.3836$ for unit 1, $z_1 = 0.4388$ and $z_2 = 1.2324$ for unit 2, and $z_1 = 0.366$ and $z_2 = 0.784$ for unit 3; Case 3: $z_1 = 0.4332$ and $z_2 = 1.6756$ for unit 1, $z_1 = 0.5868$ and $z_2 = 1.3484$ for unit 2, $z_1 = 0.428$ and $z_2 = 1.188$ for unit 3, and $z_1 = 0.21$ and $z_2 = 1.438$ for unit 4). (a) $|S_{21}|$ and $|S_{11}|$ responses (Case 1: 3-dB BW of 57% from $0.715f_0$ to $1.285f_0$; Case 2: 3-dB BW of 46.8% from $0.766f_0$ to $1.234f_0$; Case 3: 3-dB BW of 40% from $0.8f_0$ to $1.2f_0$). (b) τ_g responses (Case 1: flat-group-delay BW of 73.2% with maximum variation of ± 0.0184 ns from $0.634f_0$ to $1.366f_0$ and $\tau_g(f_0) = 0.8212$ ns; Case 2: flat-group-delay BW of 66% with maximum variation of ± 0.029 ns from $0.67f_0$ to $1.33f_0$ and $\tau_g(f_0) = 1.435$ ns; Case 3: flat-group-delay BW of 65.2% with maximum variation of ± 0.05 ns from $0.674f_0$ to $1.326f_0$ and $\tau_g(f_0) = 2.036$ ns).

higher, although the stopband-power-rejection levels are increased. In addition, their flat-group-delay ranges with lower maximum variation are slightly broadened, and they fully cover the corresponding 3-dB-BW passbands. On the other hand, examples of frequency responses of the proposed class of input-reflectionless BPFs with TZs using the resistively-terminated multi-section TL for various numbers of in-series-cascaded TL sections—i.e., $i = 1, 2$, and 3 in Fig. 10—are shown in Fig. 14. As can be seen, their 3-dB-passband BWs and stopband-power-attenuation profiles are almost unchanged, while it seems that the in-band input-power-matching levels can be increased by using a resistively-terminated TL section with more cascaded TLs. It is also observed that the maximum in-band group-delay variations are further reduced as a higher number of cascaded TLs is used in the shunt lossy TL section. This leads to a more-flattened in-band group-delay response, while their stopband group-delay shapes are almost unchanged.

As experimental demonstration, an engineered two-layer input-reflectionless wideband flat-group-delay BPF with TZs using a shunt resistively-terminated two-section TL for $i = 1$ is has been designed, simulated, fabricated, and measured. Its relevant 3-D layout is depicted in Fig. 15. Similarly, its initial design impedances associated with the TL equivalent circuit

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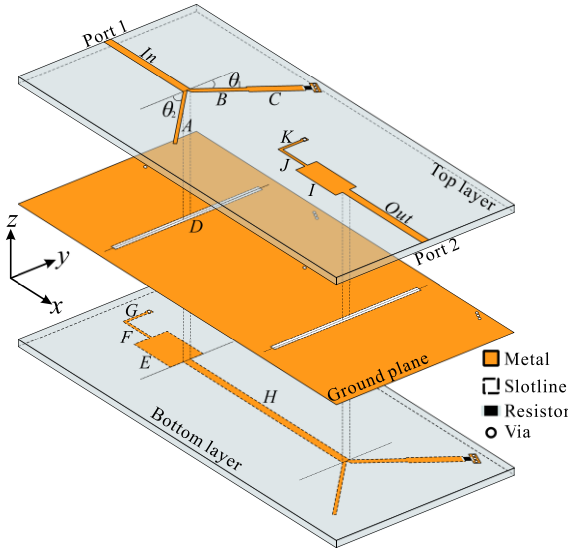


Fig. 20. 3-D layout of the developed high-order input-reflectionless wideband flat-group-delay multi-TZ BPF prototype based on two in-series-cascaded BPF units as in Fig. 10 using shunt resistively-terminated two-section microstrip line for $i = 1$ ($L_{in} = L_{out} = W_H = 40$, $L_A = 21.95$, $L_B = 21.51$, $L_C = 21.56$, $L_D = 49.7$, $L_E = 19.72$, $L_F = 13.25$, $L_G = 9.5$, $L_H = 65.61$, $L_I = 20.38$, $L_J = 12.01$, $L_K = 10$, $W_{in} = W_{out} = 1.82$, $W_A = 0.96$, $W_B = 1.88$, $W_C = 1.8$, $W_D = 0.41$, $W_E = 13.95$, $W_F = W_G = 0.67$, $W_I = 5.93$, $W_J = W_K = 0.62$ (unit: mm), and $\theta_1 = \theta_2 = 45^\circ$).

for $i = 1$ are $Z'_m = 80.55 \Omega$, $Z'_s = 67.95 \Omega$, $Z'_1 = 17.98 \Omega$, $Z'_2 = 108.08 \Omega$, $Z'_{m1} = 69.79 \Omega$, and $Z_{m2} = 52.24 \Omega$. To make the EM simulated results fairly close to the ideal ones, the simulated impedances are modified to $Z_m = 80.4473 \Omega$, $Z_s = 94.249 \Omega$, $Z_1 = 17.9706 \Omega$, $Z_2 = 111.606 \Omega$, $Z_{m1} = 62.1091 \Omega$, and $Z_{m2} = 52.3144 \Omega$. In this manner, the turns ratios of the employed transformers are extracted as $N_m = 1.0006$, $N_s = 0.8491$, $N_{m1} = 1.06$, and $N_{m2} = 1.0002$. Besides, a SMD resistor with measured resistance of 50.1Ω was soldered in the assembled BPF prototype. Fig. 16 represents the EM-simulated and measured frequency responses of the manufactured BPF prototype, along with its top- and bottom-view photographs. As can be seen in Fig. 16(a), a close agreement between the EM-simulated and measured power transmission and input-reflection responses is obtained. Due to the unwanted cross-coupling effect between the open-circuit-ended microstrip stub and the shunt resistively-terminated microstrip line—as well as to the fabrication tolerances and the discontinuities of the soldered RF SMA connectors—, the measured input-power-matching levels at the high-frequency passband region and the upper stopband are reduced, but still above 10 dB. The main performance metrics of the measured BPF circuit with TZs are the following ones: center frequency of 1.981 GHz, 3-dB-BW passband range from 1.207 to 2.754 GHz—i.e., equal to 78.09% in relative terms—, minimum in-band power-insertion-loss and input-power-matching levels equal to of 0.44 dB and 10.63 dB, respectively, and stopband-power-attenuation levels higher than 14.03 dB from DC to 5 GHz. Fig. 16(b) depicts the comparison between the simulated and measured wideband group-delay responses. In this case, the measured average in-band group-delay value is 0.6333 ns, whereas the measured maximum quasi-flat-group-delay

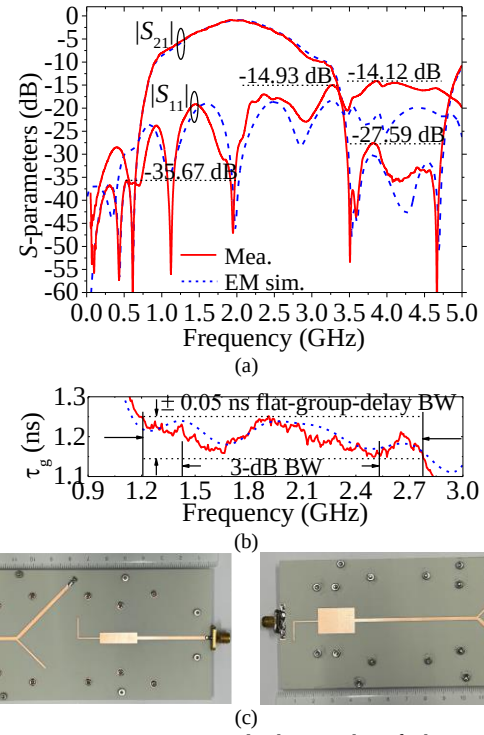


Fig. 21. Frequency responses and photographs of the manufactured microstrip prototype of the developed high-order input-reflectionless wideband flat-group-delay multi-TZ BPF in Fig. 20. (a) EM-simulated and measured power transmission ($|S_{21}|$) and input-reflection ($|S_{11}|$) responses. (b) EM-simulated and measured group-delay (τ_g) responses. (c) Top- and bottom-view photographs of the assembled prototype.

variation referred to such value is within ± 0.05 ns from 1.164 to 2.766 GHz—i.e., 81.53% in percentage terms—, thus beyond its associated 3-dB-BW passband range. The physical size of this assembled BPF prototype, whose top- and bottom-view photographs are given in Fig. 16(c), are 80×69.93 mm².

B High-Order Multi-TZ Wideband BPF With Flat Group Delay

In order to improve the performance of the type of input-reflectionless wideband flat-group-delay BPFs with TZs designed in Section III.A, its high-order versions equipped with additional pairs of close-to-passband TZs and increased stopband-power-attenuation levels are now explored. Fig. 17 shows the conceptual block diagram of the suggested high-order wideband BPF network. As shown, it is built with n ($n = 2, 3, \dots, N^*$) in-series-cascaded BPF units with TZs. Herein, these units are represented by the proposed BPF with TZs in Fig. 10 using a shunt resistively-ended two-section TL, and they are realized with identical impedances z_m , z_s , z_{m1} , and z_{m2} but distinct z_1 and z_2 values in their constituent short-circuit-ended two-section stubs. In addition, for the connection of the adjacent BPF units in the practical design, a three-quarter-wavelength-long—i.e., 3θ electrical length—50- Ω TL is used.

Based on the design procedure followed for the BPF unit with TZs in Fig. 15, a high-order input-reflectionless wideband flat-group-delay BPF with more TZs—i.e., $n = 2$ —is addressed. To forcedly meet the design specifications in (6) and (7) and obtain stopband-power-attenuation levels above

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TABLE I
COMPARISON WITH OTHER PRIOR-ART REFLECTIONLESS FLAT-GROUP-DELAY RF BPFs

Ref.	3-dB FBW/ f_0 (GHz)	In-band IL/RL (dB)	FGD FBW/MV	Stopband PAL	RR (GHz) /RL (dB)	RB	Close-to-band TZs	Design structure
[22] BPF in Fig. 16(b)	5.7%/3.5	$\leq 2.1/\geq 14.8$	7.72%/ ≤ 0.36 ns	N/A	2–6/ ≥ 8	Two-port	No	Planar
[22] BPF in Fig. 18(b)	6.72%/3.5	$\leq 1.96/\geq 15.65$	11.06%/ ≤ 0.46 ns	N/A	2–6/ ≥ 13.4	Two-port	No	Planar
[23] BPF in Fig. 5(a)	50%/2	$\leq 1^*/\geq 13^*$	62.25%/ ≤ 0.18 ns	N/A	0.6–3.5/ ≥ 12	Two-port	No	Planar
[24] BPF in Fig. 14(a)	22.8%/2.45	$\leq 2.46/\geq 11.96$	24.49%/ $\leq \pm 0.04$ ns	$\geq 12.5^*$	1.4–3.1/ ≥ 10	Two-port	Yes	Planar
[24] BPF in Fig. 14(b)	15.7%/2.51	$\leq 4.55/\geq 12.18$	23.9%/ $\leq \pm 0.04$ ns	$\geq 25^*$	1.3–3.1/ ≥ 10	Two-port	Yes	Planar
[25] BPF in Fig. 3	10.2%/5.03	$\leq 5.65/\geq 11.5$	10.5%/ $\leq \pm 0.02$ ns	$\geq 40^*$	DC–10/ ≥ 9.1	Two-port	Yes	Planar
BPF in Fig. 9	84.76%/1.956	$\leq 0.42/\geq 13.16$	96.84%/ $\leq \pm 0.05$ ns	N/A	DC–5/ ≥ 13.03	Two-port	No	Two-layer
BPF in Fig. 16	78.09%/1.981	$\leq 0.44/\geq 10.63$	81.53%/ $\leq \pm 0.05$ ns	≥ 14.03	DC–5/ ≥ 10.63	Input	Yes	Two-layer
BPF in Fig. 21	57.34%/1.9865	$\leq 0.84/\geq 14.93$	78.43%/ $\leq \pm 0.05$ ns	≥ 27.59	DC–5/ ≥ 14.12	Input	Yes	Two-layer

FBW: fractional bandwidth; IL: insertion loss; RL: return loss; FGD: flat-group-delay; MV: maximum variation; PAL: power-attenuation level; RR: reflectionless range; RB: reflectionless behavior; *: estimated from graphs.

35 dB, the design impedances of the proposed high-order BPF with TZs are properly determined. Fig. 18 represents its theoretical frequency responses. Compared with the single-unit BPF responses in Fig. 11, this high-order wideband BPF features a visibly-improved sharp-rejection stopband with four close-to-passband TZs, out-of-band-power-rejection levels above 38.2 dB, and input-power-matching levels higher than 24.72 dB within $[0, 2f_0]$. Its wideband flat group delay exhibits a maximum in-band variation of ± 0.0287 ns from $0.561f_0$ to $1.439f_0$ —i.e., 87.8% in relative terms—with regard to the group-delay value $\tau_g(f_0) = 0.7974$ ns, which covers the associated 3-dB-BW passband range from $0.712f_0$ to $1.288f_0$ —i.e., 57.6% in percentage terms. Furthermore, the frequency responses of the proposed class of high-order wideband BPFs designed with multiple in-series-cascaded BPF units—i.e., for $n = 2, 3$, and 4 , respectively—are compared in Fig. 19. As shown, by cascading more BPF units, a higher-order wideband BPF with narrowed 3-dB BW range, more pairs of close-to-passband TZs, and highly-enhanced filtering selectivity and stopband-power-attenuation levels [35] is attained. However, the input-power-matching levels are slightly deteriorated as a higher number of BPF units is used. Besides, for more BPF units and their inter-connection lines, the group-delay value $\tau(f_0)$ and maximum group-delay variation are increased, while a narrowed flat-group-delay BW is correspondingly obtained.

To verify the above-suggested high-order wideband BPF concept, a two-layer microstrip prototype using two in-series-cascaded input-reflectionless BPF units as in Fig. 10 has been designed, simulated, implemented, and tested. Fig. 20 depicts the resulting 3-D layout structure. It is initially realized with $Z'_m = 70.3 \Omega$, $Z'_s = 62.7 \Omega$, $Z'_{m1} = 48.24 \Omega$, $Z_{m2} = 49.69 \Omega$, and $Z'_1 = 10.23 \Omega$, $Z_2 = 83.07 \Omega$ for unit 1, and $Z'_1 = 21.14 \Omega$, $Z_2 = 86 \Omega$ for unit 2. However, in the practical design, impedance parameters as $Z_m = 70.3034 \Omega$, $Z_s = 91.4682 \Omega$, $Z_{m1} = 48.3161 \Omega$, $Z_{m2} = 49.6309 \Omega$, and $Z_1 = 10.2283 \Omega$, $Z_2 =$

83.0158Ω for unit 1, and $Z_1 = 21.1518 \Omega$ and $Z_2 = 85.7904 \Omega$ for unit 2 were used to make the EM-simulation results closely meet the theoretical ones. In addition, two SMD resistors with measured resistances of 50.2Ω and 50.1Ω were employed. The EM-simulated and measured frequency responses and the photographs of the manufactured high-order BPF prototype are provided in Fig. 21. Again, as observed in Fig. 21(a) and (b), a fairly-close agreement between the EM simulations and measurement is obtained. The main measured performance metrics of the high-order BPF prototype with four close-to-passband TZs are summarized as follows: center frequency of 1.9865 GHz, 3-dB-bandwidth frequency range from 1.417 to 2.556 GHz—i.e., 57.34% in relative terms—, minimum in-band power-insertion-loss level and stopband-power-attenuation level of 0.8411 dB and 27.59 dB, respectively, and input-power-matching levels above 14.12 dB from DC to 5 GHz. Besides, the measured maximum group-delay variation is within ± 0.05 ns from 1.213 to 2.778 GHz—i.e., 78.43% in percentage terms—in relation to the measured in-band average group-delay value of 1.178 ns, which is larger than the measured 3-dB-BW passband range. The top- and bottom-view photographs of the assembled BPF prototype with a circuit size of $145.61 \times 69.93 \text{ mm}^2$ are depicted in Fig. 21(c).

IV. COMPARISON WITH THE STATE-OF-THE-ART DESIGNS

In order to quantitatively demonstrate the superiorities of the developed two-port- and input-reflectionless wideband flat-group-delay RF BPF prototypes of this work, a comparison between them and other prior-art absorptive BPFs with flat-group-delay responses is given in Table I. As can be seen, the reflectionless wideband flat-group-delay BPF circuits of this work are featured with the expected superior properties in terms of broader 3-dB BW and associated flat-group-delay BW with regard to a remarkably-small maximum in-band group-delay variation—which covers the 3-dB-BW passband in all cases—, along with wideband absorptive spectral range.

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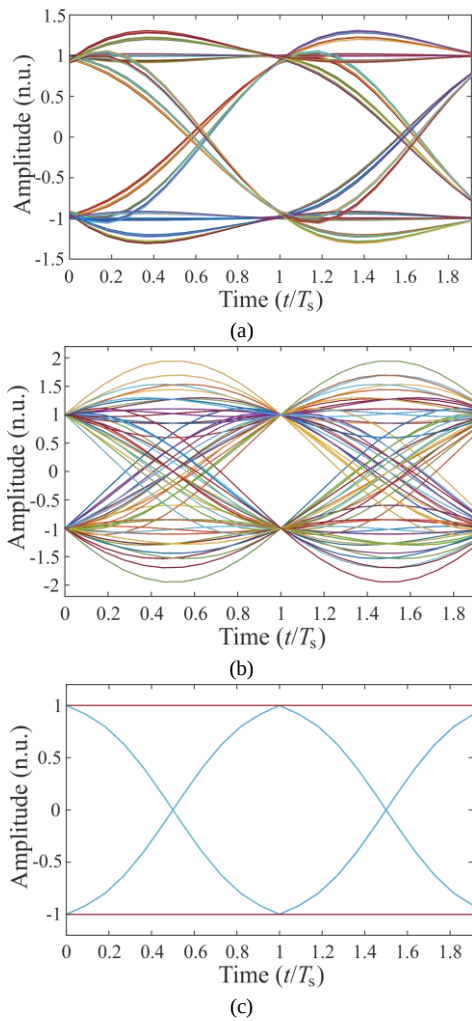


Fig. 22. Simulated eye diagrams for a binary communications system (T_s is the symbol period and eleven samples per symbol have been considered). (a) Non-flat group-delay filter. (b) Raised-cosine filter. (c) Flat group delay filter.

V. SUMMARY AND CONCLUDING REMARKS

A class of two-layer reflectionless microstrip RF BPFs with wideband flat-group-delay responses has been reported. By means of a microstrip-to-microstrip vertical transition and two shunt resistively-terminated one-section microstrip lines, the properties of two-port-reflectionless wideband BPFs with low-in-band-variation equi-ripple-type group-delay characteristics have been firstly determined. Then, its wideband extremely-flat-group-delay BPF counterpart has been conceived by exploiting shunt resistively-ended two-section microstrip lines. Besides, by using a modified microstrip vertical transition with a short-circuit-ended two-section microstrip line at the output port and a shunt lossy two-section microstrip line, a type of input-reflectionless wideband BPF with a pair of TZs and flat-group-delay profile has been designed. By cascading two replicas of such BPF unit with two TZs, a novel kind of high-order wideband flat-group-delay BPF with more pairs of close-to-passband TZs and increased out-of-band-attenuation

levels has been discussed. The RF operational principles and theoretical foundations of these engineered wideband BPFs have been presented. Moreover, three experimental proof-of-concept prototypes of the engineered two-layer absorptive wideband flat-group-delay BPF devices have been designed, simulated, constructed, and tested. The fairly-close agreement obtained between their EM-simulated and measured results has validated the devised design concept. This family of RF BPFs, which show wideband quasi-flat group-delay responses and broadband absorptive behavior, are expected to find application in modern high-data-rate digital-communication systems with highly-stringent phase-distortion requisites. Further research work to be carried out is the development of a complete high-data-rate digital-communication system and the practical evaluation of the developed RF BPFs into it for RF signals employing different digital-modulation schemes (e.g., quadrature amplitude or phase-shift keying modulations).

APPENDIX A

To illustrate the benefits referred in Section I derived from the use of flat-group-delay filters to process communications signals in scenarios with stringent phase-distortion requisites, the simulated eye diagrams for a binary communications system example are shown in Fig. 22. Specifically, three cases consisting of employing a non-flat-group-delay filter, a raised-cosine filter, and a flat-group-delay filter are contemplated.

Fig. 22(a) shows a simulated eye diagram corresponding to the output signal of a conventional filter with a non-flat group-delay profile—i.e., with remarkable variations or peaks in the transition bands. As can be seen, there is no sampling instant for which the ISI phenomenon is circumvented. Conversely, Fig. 22(b) shows the eye diagram for the output signal of a raised-cosine filter. In this case, there is an optimum sampling time—i.e., the center of the eye—for which there is no ISI. However, small jitter errors would lead to undesired ISI. This jitter can be more apparent for high data rates as in wideband digital-communications scenarios, which intrinsically involve small symbol periods. This referred intolerance to jitter is easily observed in the zero-crossing intervals of the eye diagram in Fig. 22(b). Contrary to that, a flat group-delay filter processing communications signals does not add ISI effects at any instant in the symbol period, thus being fully tolerant to possibly arising jitter errors. This is demonstrated in Fig. 22(c), which attests the benefits of using this type of filter in terms of ISI despite its rounded-shaped transmission band in amplitude.

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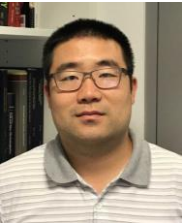
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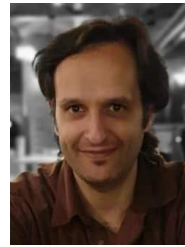
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